

Road Accident Risk Predictor — SSD 2026 Project Showcase Submission

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EXPLAINABLE RISK SCORING FOR INFRASTRUCTURE SAFETY PRIORITIZATION

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[Submitted to SSD 2026 — Project Showcase, track “AI, Data & Smart Systems for Sustainability”](#)

Summary

[Machine learning identifies which road conditions carry disproportionate accident risk, so limited infrastructure safety spending can be targeted rather than spread evenly — with SHAP explanations for every prediction.](#)

Problem

[Road safety agencies cannot treat every road segment equally with finite inspection and engineering resources. Road trauma is concentrated in particular conditions rather than spread evenly across a network. The challenge is identifying which conditions carry disproportionate risk so intervention — signage, lighting, speed-limit review — can be prioritized where it has the most potential benefit.](#)

Solution

[A Random Forest Regressor trained on 517,754 road-condition scenarios \(15 features, including three engineered interaction terms\) predicts a risk score for a given road/environment combination and explains *why* via SHAP feature attribution — not just a number, but which factors are driving it \(speed, curvature, lighting, weather\). Live and publicly deployed at the link above.](#)

Evidence of impact

Ranking all training rows by predicted risk: the **top 10% of road conditions carry 1.93x their proportional share of total predicted risk** (19.3% of total risk mass concentrated in 10% of segments). This is measured directly from the model's own output, not assumed.

<u>Risk tier</u>	<u>% of segments</u>	<u>Share of total risk mass</u>	<u>Concentration ratio</u>
<u>Top 1%</u>	<u>1%</u>	<u>2.5%</u>	<u>2.50x</u>
<u>Top 5%</u>	<u>5%</u>	<u>11.1%</u>	<u>2.22x</u>
<u>Top 10%</u>	<u>10%</u>	<u>19.3%</u>	<u>1.93x</u>
<u>Top 25%</u>	<u>25%</u>	<u>41.0%</u>	<u>1.64x</u>

Under scenario-based intervention-effectiveness assumptions (15–30%, sourced from FHWA's Crash Modification Factors Clearinghouse — a repository of 3,000+ peer-reviewed countermeasure studies), this represents a targeting mechanism with measurable potential benefit. **This is stated as a scenario-based potential impact, not a causal claim** — the model has not been shown to have prevented any crash. Full methodology and worked calculation: [IMPACT.md](#) in the supporting files.

SDG alignment

Primary — SDG 3.6: halve global deaths and injuries from road traffic accidents. **Secondary — SDG 11.2:** safe, affordable, sustainable transport systems.

Mechanism: risk scoring → targeted infrastructure spend → fewer serious injuries. The model does not predict who will crash — it identifies which road conditions carry disproportionate predicted risk.

What is innovative

1. **Predictive** — moves from historical reporting toward forward-looking risk prioritization.
2. **Explainable** — SHAP translates every prediction into interpretable contributing factors, not a black box.
3. **Action-oriented** — connects risk scores to infrastructure intervention prioritization, not just a number on a screen.

Technical evidence

[Random Forest Regressor, 200 trees, 517,754 training rows, validation \$R^2\$ 0.88 — independently reproduced \(0.8835\), not just claimed. 5-fold CV std \$\pm 0.0003\$. No data leakage: no duplicate IDs, and the dataset has no geographic or temporal columns for a random split to leak across. Subgroup performance checked by road type and weather \(stable, \$R^2\$ 0.877–0.885\) and by lighting \(small honest degradation toward night, \$R^2\$ 0.857→0.846 — disclosed, not hidden\). Full detail: \[MODEL_VALIDATION.md\]\(#\) .](#)

Responsible AI

[The model uses road and environmental conditions only — no driver identity, ethnicity, or demographic data. Intended for infrastructure planning and safety prioritization, explicitly **not** for individual driver liability, profiling, or enforcement. This is a live decision- support prototype, not a production system with an operator or SLA. Known limitations, including a disclosed subgroup- performance gap in low-light conditions and the boundary of what this synthetic \(Kaggle Playground Series S5E10\) dataset can support, are documented in full in \[RESPONSIBLE_AI.md\]\(#\) .](#)

Live demo

https://roadaccident-roshana-aryal.streamlit.app/SMARTPANTS_PRESERVED_151SMARTPANTS_PRESERVED_152SMARTPANTS_PRESERVED_153

[Hosted on Streamlit Community Cloud’s free tier, which sleeps after inactivity — cold start measured at ~95 seconds. If the link shows “Zzzz... this app has gone to sleep,” click “Yes, get this app back up!” and wait roughly a minute. Screenshots and a walk-through recording are included in the supporting files as a fallback.](#)

Contents of the supporting zip

- [This document \(\[PROJECT_SUBMISSION.pdf\]\(#\) \)](#)
- [screenshots/](#) — [live gauge prediction with SHAP contributions, model info and feature importance](#)
- [walkthrough.mp4](#) — [60–90s recorded demo \(if included\)](#)
- [IMPACT.md](#) , [RESPONSIBLE_AI.md](#) , [MODEL_VALIDATION.md](#) — [full supporting detail behind the summaries above](#)